

Reproduction of valkey-io/valkey: valkey-benchmark throughput

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Independent reproduction · Benchmark or leaderboard entry · Databases

Abstract

Built Valkey from source at the pinned commit (make, default settings, jemalloc) and ran the bundled valkey-benchmark against a local server with persistence disabled: 2,000,000 requests, 50 parallel clients, pipeline depth 16, for SET, GET, LPUSH and LPOP, reporting requests per second and latency percentiles. Server and benchmark ran on the same machine (Intel Xeon W-2145, 16 threads).

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

This cover page is not signed. Check it against the SHA-256 listed at <https://kveritas.org/records/2610.00023v1>, and verify each report with its code bundle: `kveritas verify <report.pdf> --bundle <bundle.zip>`.

1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	valkey-benchmark SET/GET/...	2026-10-08	1	1	Signed by the K-Veritas server

2 Results

r1: valkey-benchmark SET/GET/LPUSH/LPOP

Run	Metric	Final value	Step
1	set_rps	826788	
1	set_p50_ms	0.743	
1	set_p99_ms	1.927	
1	get_rps	1.00705e+06	
1	get_p50_ms	0.567	
1	get_p99_ms	1.879	
1	lpush_rps	928505	
1	lpush_p50_ms	0.671	
1	lpush_p99_ms	1.903	
1	lpop_rps	870322	
1	lpop_p50_ms	0.743	
1	lpop_p99_ms	1.687	

Run	Inline claim	Value	Phase
1	set_rps	826788	
1	get_rps	1.00705e+06	
1	lpush_rps	928505	
1	lpop_rps	870322	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	9.6 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	d1488025e31ef7123382d9fb0ed02f96379ff61a34eac204fd663e57818c3102	1.5 MB
r1.kvbundle.zip	e0376bad0b10051f8f11be543456db792e21942d5d133c9046d6df57e30fcd19	3.1 MB

5 Links and licence

Reproduction of: <https://github.com/valkey-io/valkey/tree/ea4d4e7d561895f64a86238bc587063c79c23da7>

Record: CC-BY 4.0. Code: BSD-3-Clause, as declared by the authors. Tags: key-value-store, valkey, benchmark, throughput.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (valkey-benchmark SET/GET/LPUSH/LPOP), kv:2610.00023v1/r1. Data hash 736e1b97ce7768fcc5c3e14f4ead842280fcd30b6880f202c5804568d9bce3d2.

<https://kveritas.org/records/2610.00023v1/r1.pdf>